

CAMERON FALLS, ON, Canada

WMO: 712910

Lat: 49.154N

Lon: 88.345W

Elev: 763

StdP: 14.30

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.8	-16.4	-26.5	1.3	-20.4	-22.0	1.7	-15.6	18.8	6.5	16.9	7.6	2.3	20	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	81.9	66.7	78.8	64.9	75.9	63.2	69.9	78.0	67.4	74.5	65.4	72.4	6.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.9	101.9	74.4	64.8	94.6	71.4	62.8	87.9	68.5	34.4	78.2	32.3	74.3	30.7	72.4	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	15.5	13.6	-26.5	88.2	5.1	3.2	-30.1	90.4	-33.1	92.3	-36.0	94.1	-39.7	96.4		
(5)			-26.7	73.6	5.0	2.9	-30.3	75.7	-33.3	77.4	-36.1	79.1	-39.8	81.2	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.0	6.8	8.8	21.2	34.8	46.8	57.4	63.6	62.9	54.7	42.1	28.8	13.7	(6)	(7)
	DBStd	22.02	13.12	11.66	12.12	8.20	7.14	5.49	5.14	5.58	7.37	7.70	10.31	12.45	(7)	(8)
	HDD50	6067	1340	1154	892	456	151	7	0	0	36	270	637	1124	(8)	(9)
	HDD65	10340	1805	1574	1357	905	566	233	88	109	317	710	1087	1589	(9)	(10)
	CDD50	1304	0	0	0	1	51	230	422	399	176	25	0	0	(10)	(11)
	CDD65	101	0	0	0	0	0	6	45	42	7	1	0	0	(11)	(12)
	CDH74	1060	0	0	0	0	27	121	507	349	54	2	0	0	(12)	(13)
	CDH80	187	0	0	0	0	1	17	101	59	8	1	0	0	(13)	(14)
(14) Wind	WSAvg	5.2	4.9	5.3	5.9	6.2	6.0	5.0	4.6	4.5	4.8	5.2	5.1	4.8	(14)	(15)
(15) Precipitation	PrecAvg	34.1	2.1	1.3	1.9	1.8	2.6	3.7	3.5	4.1	3.9	3.6	3.1	2.5	(15)	(16)
	PrecMax	48.9	4.6	3.2	6.0	4.1	5.4	8.7	6.8	8.0	8.5	9.4	6.2	5.4	(16)	(17)
	PrecMin	23.9	0.4	0.2	0.3	0.3	0.4	1.4	1.5	1.6	1.7	1.2	1.0	1.0	(17)	(18)
	PrecStd	5.9	1.1	0.7	1.2	1.0	1.2	1.9	1.4	1.6	1.8	1.9	1.3	1.1	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.9	39.9	55.9	63.8	77.8	82.3	86.9	85.7	79.7	69.3	54.6	40.1	(19)	(20)
		MCWB	34.8	34.8	45.6	48.2	58.4	65.4	68.8	69.3	65.3	57.9	48.9	38.1	(20)	(21)
	2%	DB	34.1	35.2	46.8	57.1	71.9	77.2	82.9	81.0	74.4	63.1	49.1	37.2	(21)	(22)
		MCWB	33.0	32.7	40.2	44.3	55.9	61.4	66.8	67.1	63.7	54.6	45.4	35.6	(22)	(23)
	5%	DB	31.9	31.4	40.9	52.8	66.8	74.0	79.6	77.9	70.1	58.3	45.7	34.7	(23)	(24)
		MCWB	30.9	29.3	35.4	42.1	53.2	60.0	65.6	65.3	61.6	52.5	42.6	33.4	(24)	(25)
	10%	DB	27.6	27.5	37.3	48.8	61.8	70.7	76.3	74.7	66.4	54.6	42.6	32.2	(25)	(26)
		MCWB	26.3	25.5	33.8	40.1	50.9	58.0	64.0	63.9	60.1	50.4	40.3	31.1	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.6	36.3	47.4	49.8	62.6	68.4	72.7	73.4	68.8	60.7	50.2	38.7	(27)	(28)
		MCDB	36.5	38.2	52.5	58.8	69.6	76.0	81.9	83.4	75.1	65.0	53.5	39.5	(28)	(29)
	2%	WB	33.2	32.8	40.4	45.9	58.1	64.3	69.9	70.2	65.7	56.5	46.0	35.9	(29)	(30)
		MCDB	33.9	35.0	45.2	54.7	66.6	72.5	79.3	77.7	70.9	60.6	48.4	36.8	(30)	(31)
	5%	WB	30.7	29.4	36.7	43.1	55.2	62.0	67.2	67.8	63.4	53.4	43.0	33.6	(31)	(32)
		MCDB	31.9	31.2	39.9	50.4	64.3	70.3	75.6	74.1	67.9	57.2	45.0	34.4	(32)	(33)
	10%	WB	26.3	25.5	34.1	40.7	52.5	60.0	65.2	65.6	61.3	50.5	40.3	31.0	(33)	(34)
		MCDB	27.5	27.3	36.9	47.7	60.2	67.4	73.3	71.7	65.3	54.3	42.6	32.3	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.5	19.9	20.3	19.1	20.9	21.4	20.9	19.7	17.9	13.5	12.1	14.3	(35)	(36)
		MCDBR	15.8	20.1	22.0	27.2	31.8	29.1	27.3	25.3	22.3	21.4	15.6	13.9	(36)	(37)
		MCWBR	14.7	17.0	15.2	15.6	16.6	14.8	13.4	13.1	13.1	14.0	11.2	12.8	(37)	(38)
	5% WB	MCDBR	15.2	18.9	18.7	24.2	24.9	23.3	23.9	21.9	18.3	18.7	14.2	12.6	(38)	(39)
		MCWBR	14.4	16.5	13.1	15.1	14.3	13.3	13.2	12.6	11.8	13.4	11.0	12.1	(39)	(40)
(40) Clear-Sky Solar Irradiance	taub		0.255	0.269	0.301	0.342	0.354	0.372	0.393	0.368	0.354	0.312	0.284	0.264	(40)	(41)
	taud		2.351	2.318	2.345	2.376	2.428	2.402	2.381	2.447	2.483	2.568	2.496	2.361	(41)	(42)
	Ebn at Noon		259	282	290	287	287	282	273	276	269	266	248	238	(42)	(43)
	Edh at Noon		21	28	32	35	35	36	36	32	28	22	18	18	(43)	(44)
(44) All-Sky Solar Radiation	RadAvg		378	734	1146	1513	1690	1813	1817	1550	1105	645	333	271	(44)	(45)
	RadStd		47	68	101	148	99	132	118	125	84	90	36	35	(45)	(46)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46) N/A
(47)	Regional (11 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(47) N/A

Nomenclature: See separate page